

Question	Answer	Marks	Guidance
4(a)	$E_{(K)} = \frac{1}{2}mv^2$	C1	Allow any value of energy equated to $\frac{1}{2}mv^2$
	$v^2 = (2 \times 1.3 \times 10^5 / 1.5 \times 10^4)$	C1	When candidates initially state the correct symbol equation, but then at the substitution stage write v instead of v^2 , treat this as TE in the 2nd mark and allow ECF to the 3rd mark (max 2/3 marks). When candidates do not initially state a symbol equation, and then at the substitution stage write v instead of v^2 , this implies an incorrect symbol equation and so treat as XP (no credit 0/3).
	$v = 4.2 \text{ m s}^{-1}$	A1	

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4(b)(i)	work (done) / time (taken)	B1	Allow 'work (done) per (unit) time' Allow 'ratio of work and time' Allow 'rate of (doing) work' Allow 'work over time' (BOD) Not 'work in unit time' Not 'rate of (doing) work per unit time' Not 'change of work (done) per unit time' Ignore 'work (done) by time' Ignore 'work (done) into time' Allow 'energy transferred / time' or 'energy converted / time' Allow 'change in energy / time' (BOD) Allow 'rate of energy transferred' Ignore just 'energy / time' Ignore symbols unless symbols are explained Ignore any units e.g. ignore 'work done per second' or 'Joules per time'

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4(b)(ii)	$W = Fs$ or $W = Fvt$ or $W = Pt$ useful work done = change in $E_p + Fs$	C1	Allow $(\Delta)E$ for W and allow 'd' or 'x' for 's' Allow 'WD' for work done (BOD) Allow any value of energy equated to Fs for 1st C1 mark
	$W = (1.1 \times 10^5 \times 52) - (13.0 \times 10^6 - 7.4 \times 10^6)$ $= 1.2 \times 10^5$	C1	Allow Fs or Fvt or Pt for W Treat not converting or incorrect conversion of both 13.0 MJ and 7.4 MJ to J as a single POT error ECF speed from 4(a)
	$F = 1.2 \times 10^5 / (4.2 \times 52)$ $= 550 \text{ N}$	A1	Allow 555 N (correct to 3SF because use of $v=4.16$ gives 554.7 N) but do not allow any other value that rounds to 560
	OR $P = Fv$ useful output power = power to raise weight + Fv	(C1)	Allow any value of power equated to Fv for this mark
	$P = 1.1 \times 10^5 - [(13.0 \times 10^6 - 7.4 \times 10^6) / 52]$ $= 2.3 \times 10^3$	(C1)	Allow Fv for P Treat not converting or incorrect conversion of both 13.0 MJ and 7.4 MJ to J as a single POT error ECF speed from 4(a)
	$F = (2.3 \times 10^3) / 4.2$ $= 550 \text{ N}$	(A1)	Allow 555 N (3SF) but Not 560 N
	OR $P = Fv$ driving force = component of weight parallel to slope + F	(C1)	

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4(b)(ii)	$1.1 \times 10^5 / 4.2 = F + 1.5 \times 10^4 \times 9.81 \sin \theta$ <u>and</u> $\sin \theta = (13.0 \times 10^6 - 7.4 \times 10^6) / (1.5 \times 10^4 \times 9.81 \times 4.2 \times 52)$ $1.1 \times 10^5 / 4.2 = F + (13.0 \times 10^6 - 7.4 \times 10^6) / (4.2 \times 52)$	(C1)	ECF speed from 4(a) Treat not converting or incorrect conversion of both 13.0 MJ and 7.4 MJ to J as a single POT error
	$F = 2.619 \times 10^4 - 2.564 \times 10^4$ $= 550 \text{ N}$	(A1)	Allow 555 N (3SF) but not 560 N
4(b)(iii)	$P = IV$ $= 95 \times 1400$ $= 1.33 \times 10^5$	C1	For both methods C1 marks may be credited independently Allow $V = IR$ <u>and</u> either $P = V^2 / R$ or $P = I^2 R$
	efficiency = useful output power / (total) input power ($\times 100$)	C1	Ignore 'useful output/total input' but allow 'power' to be implied by substituted values of power Candidates may use useful output power = $(13.0 \times 10^6 - 7.4 \times 10^6) / 52 = 1.07692 \times 10^5$ so Allow $(1.08 \times 10^5 / (95 \times 1400)) (\times 100)$ to imply both C1 marks (but will lead to incorrect final answer 81%)
	$= [1.1 \times 10^5 / (95 \times 1400)] (\times 100)$ $= 83\%$	A1	
	OR $E = VIt$ $= 95 \times 1400 \times 52$ $= 6.92 \times 10^6$	(C1)	
	efficiency = useful output energy / (total) input energy ($\times 100$)	(C1)	Ignore 'useful output/total input' but allow 'energy' to be implied by substituted values of energy Candidates may use useful output energy = $(13.0 \times 10^6 - 7.4 \times 10^6) = 5.6 \times 10^6$ so Allow $(5.6 \times 10^6 / (95 \times 1400 \times 52)) (\times 100)$ to imply both C1 marks (but will lead to incorrect final answer 81%)
	$= [(1.1 \times 10^5 \times 52) / (1400 \times 95 \times 52)] (\times 100)$ $= 83\%$	(A1)	